Master Project Final Report

First-principles studies of quantum defect candidates in 2D WS2

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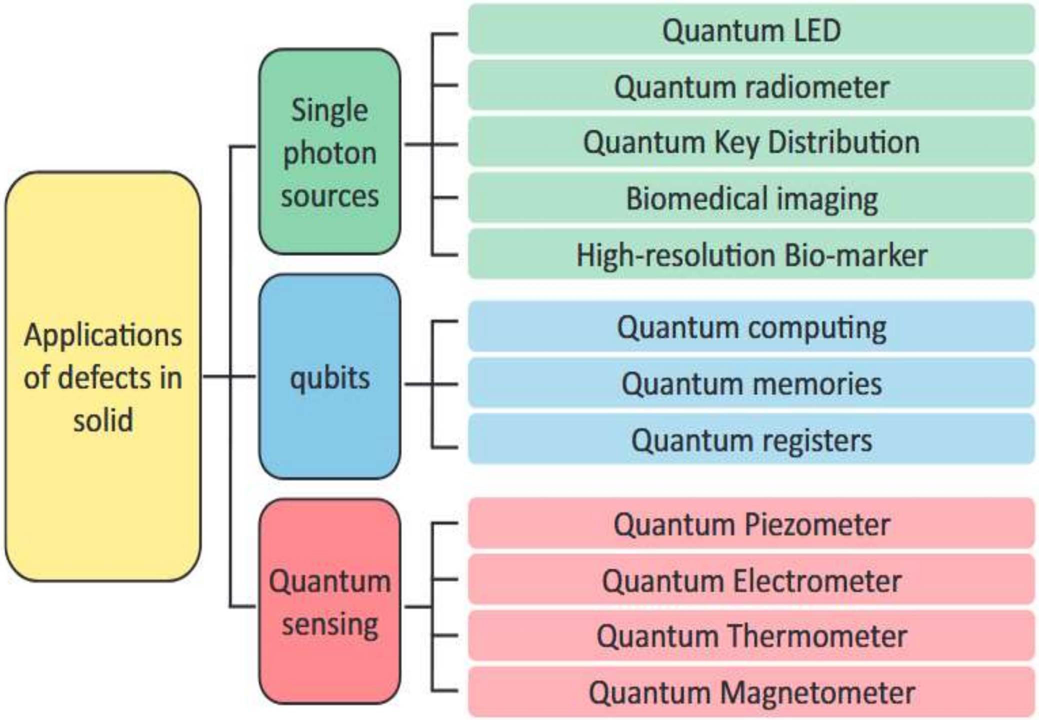
Background

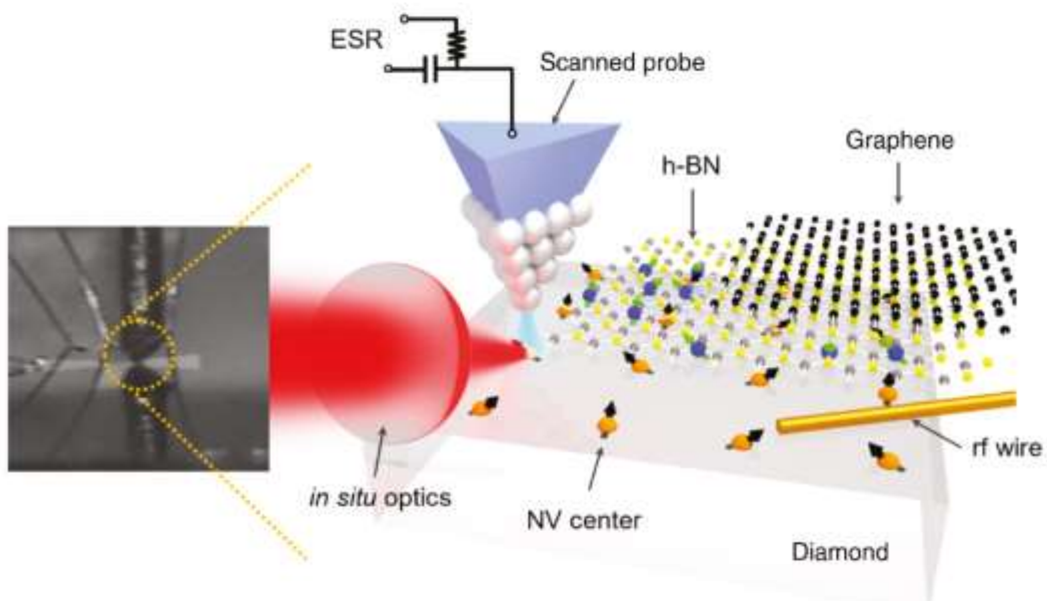
Quantum point defect (QPD)



Background

Deterministic control of quantum point defects in 2D materials

Current STM-based techniques allows for deterministic placement and control of individual QPDs. 2D host materials (Graphene, h-BN, and TMDs) are ideal platforms.



Lee, D.; Gupta, J. A. *Nanophotonics* **2019**, *8* (11), 2033–2040.

Background

Key Properties of Quantum Defects

Category	Defect properties
Basic ground-state properties	Geometries, defect formation energies, charge transition levels, spin multiplicity
Specific ground-state properties	Magnetic interactions (zero-field splitting, spin-orbit, hyperfine), vibrational modes and frequencies, electron-phonon coupling, spin-phonon coupling
Excited-state properties	Excited-state energies and geometries, multiplet structure, transition dipole moments, radiative rates, optical lineshapes, nonradiative transitions

Bassett, L. C et al. *Nanophotonics* **2019**, *8* (11), 1867–1888.

Workflow

Substitutional 4d&5d TM defects in 2D WS_2

Determine defect supercell

manually set three initial dopant positions

ground state structure
with possible symmetry distortion

Calculate formation energy

defect charge states
within pristine bandgap

Evaluate spin configuration

Singlet or triplet ground state

Bandstructure Analysis

defect level

Defect Generation

Initial Disruption Setting - relax the defect structures in symmetry broken supercells

Trigonal prismatic coordination → $\mathbf{D_{3h}}$

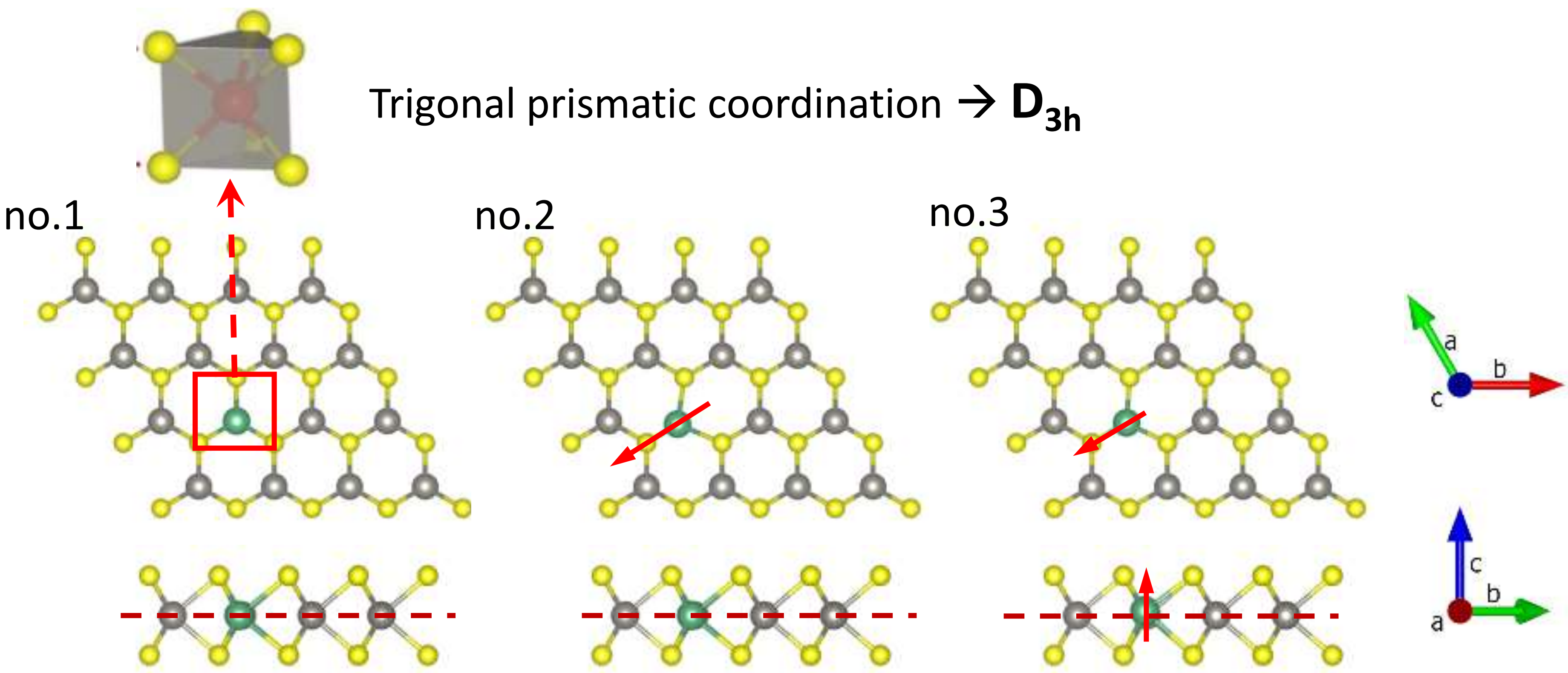
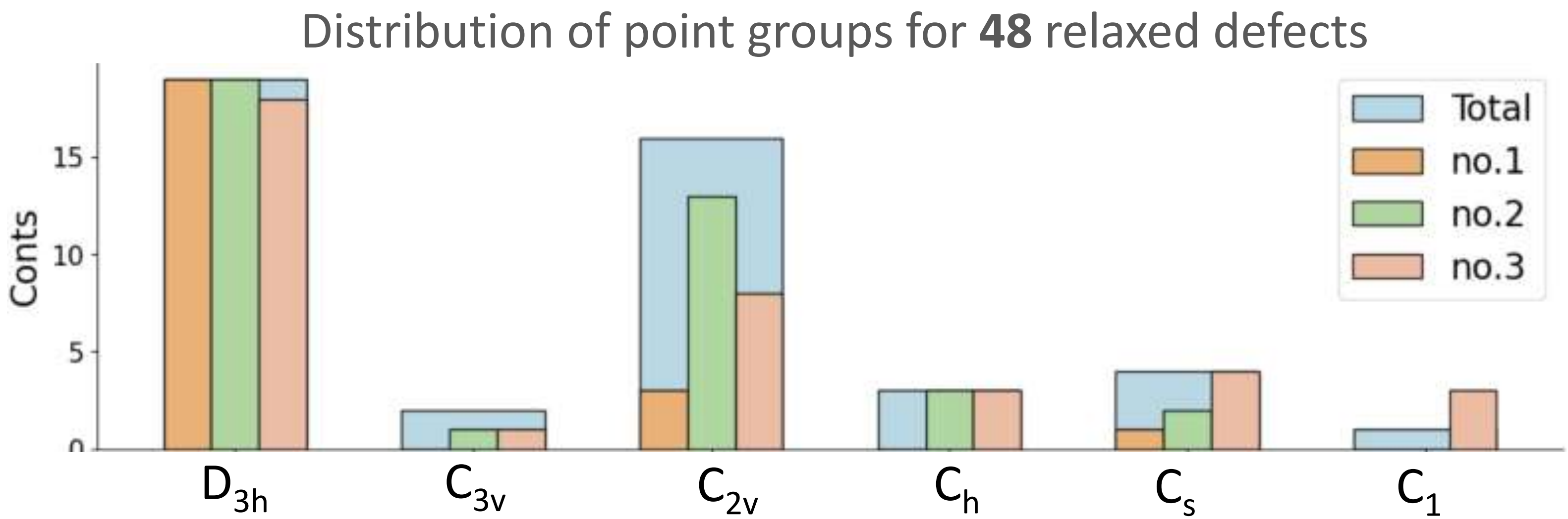
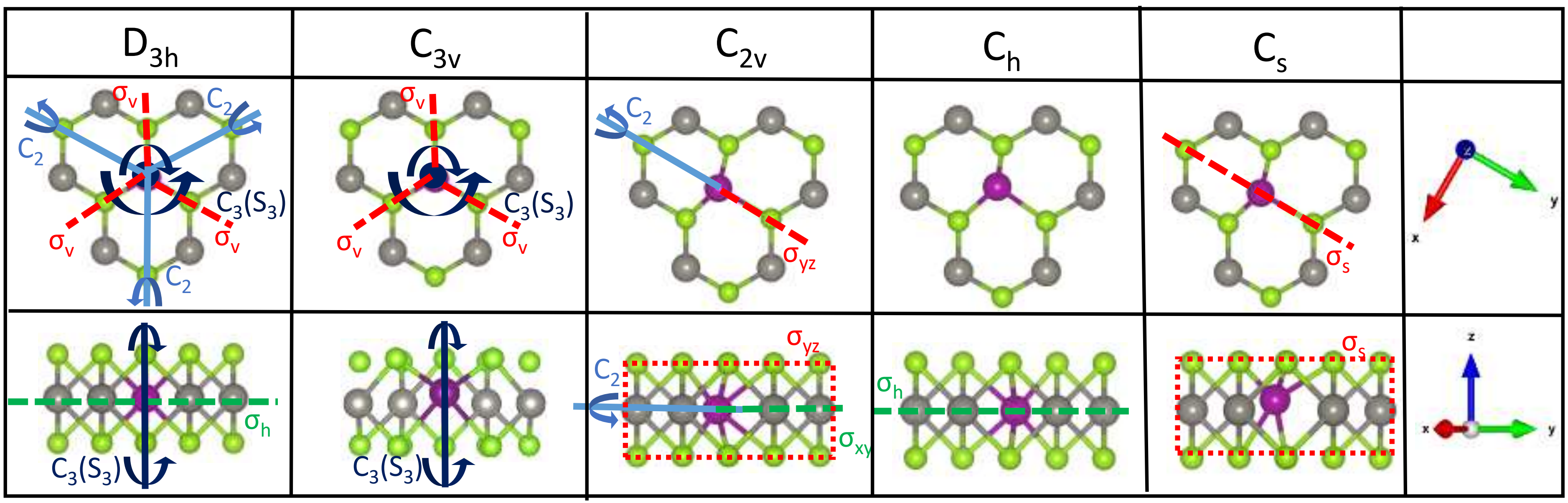


Fig.1 Top views and side views of the possible structures and symmetries

16 TM elements (4d&5d) → 48 charged defects (-1,0,+1) → 144 initial structures

Defect Relaxation





Defect Formation Energy

$$\Delta H_{\mathrm{D,q}}(E_F, \mu) = [E_{\mathrm{D,q}} - E_{\mathrm{H}}] + \sum_i n_i \mu_i + qE_F + E_{\mathrm{corr}}$$

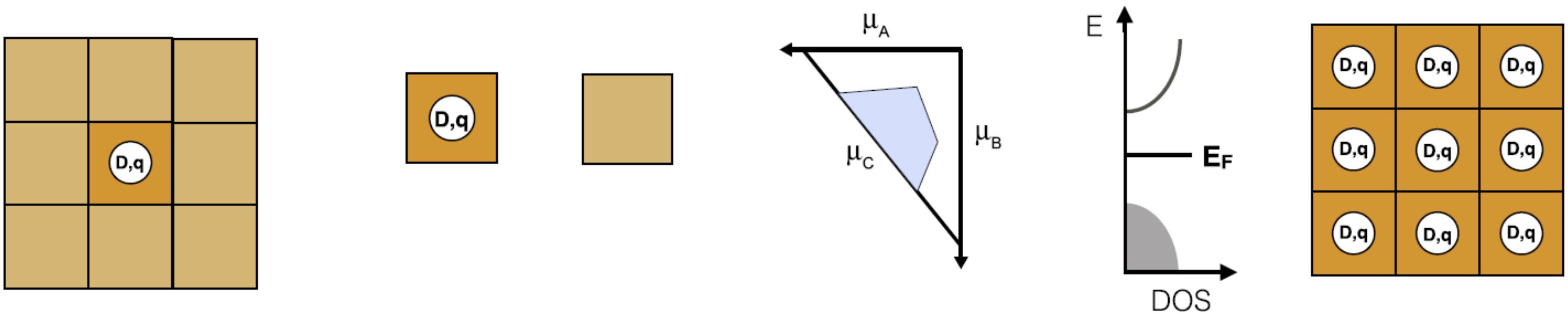
Defect Formation Energy

① Defect Supercell | Host Supercell

② Chemical potentials from phase stability

③ Electron chemical potential

④ Finite size corrections



Goyal, A. et al, *Comp. Mat. Sci.*, **2017**, *130*, 1-9

D = defect **q** = charge state **H** = host

$\mathbf{n_i}$ = number of atoms added to (negative) or removed from (positive) the host

$\boldsymbol{\mu_i}$ = chemical potential **i** = atomic species

Atom Chemical Potential

Constraints by phase stability:

$$\mu_W + 2\mu_S = E[\mathrm{WS_2}] \quad (1)$$

$$x\mu_{Nb} + y\mu_S \leq E[\mathrm{Nb}_x S_y] \quad (2)$$

$$\mu_W \leq E[\mathrm{W}] \quad (3)$$

$$\mu_S \leq E[\mathrm{S}] \quad (4)$$

$$\mu_{Nb} \leq E[\mathrm{Nb}] \quad (5)$$

$$\Delta\mu_A = \mu_A - \mu_A^0 \quad (6)$$

$$\mu_A^0 = E[A]$$

$$\Delta\mu_{\mathrm{W}} + 2\Delta\mu_{\mathrm{S}} = \Delta E[\mathrm{WS_2}] \quad (7)$$

$$x\Delta\mu_{\mathrm{Nb}} + y\Delta\mu_{\mathrm{S}} \leq \Delta E[\mathrm{Nb}_x S_y] \quad (8)$$

$$\Delta\mu_W \leq 0 \quad (9)$$

$$\Delta\mu_S \leq 0 \quad (10)$$

$$\Delta\mu_{Nb} \leq 0 \quad (11)$$

E[A]: bulk energy per formula unit

$$\Delta E[\mathrm{A}_x\mathrm{B}_y] = E[\mathrm{A}_x\mathrm{B}_y] - xE[\mathrm{A}_x] - yE[\mathrm{B}_y] \quad (12)$$

μ_A^0 In electronic-structure calculations, chemical potentials can be referenced to the total energy of the elemental phases at T = 0 K.

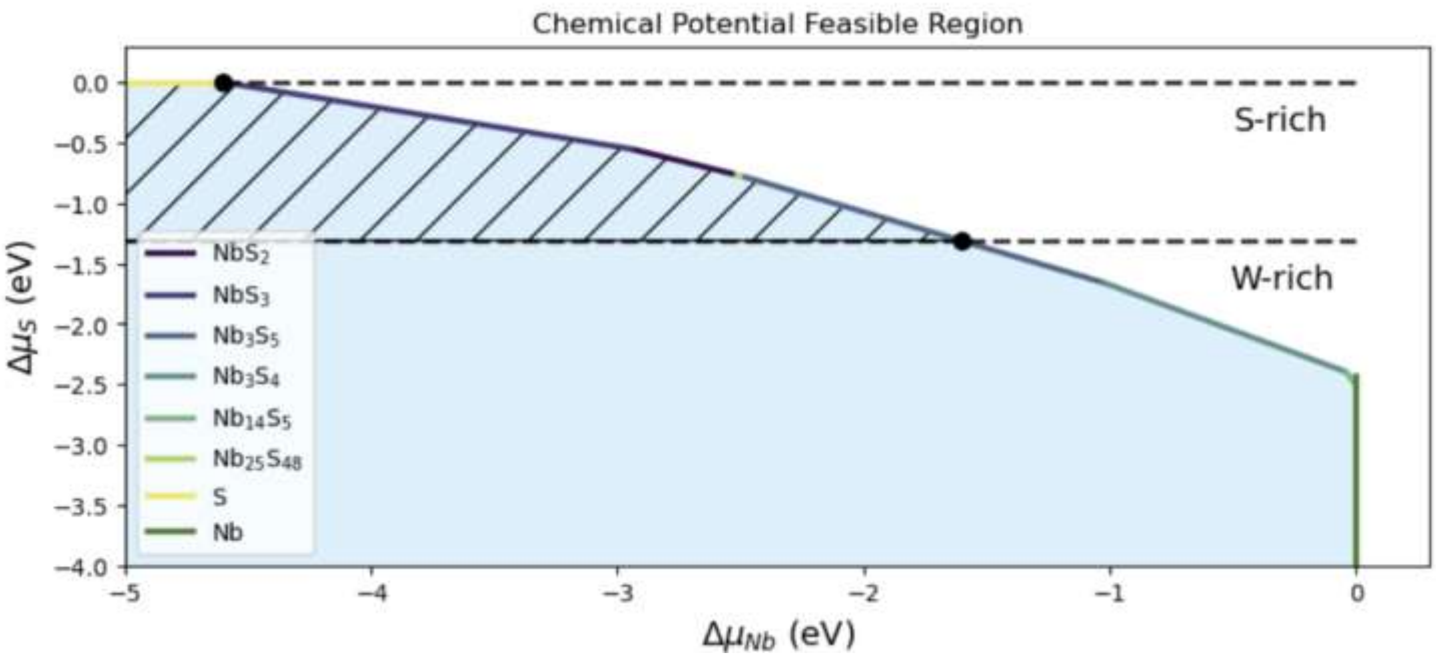
ΔE ΔH Energy enthalpy and formation energy are considered to have the same value since the pressure is small enough after relaxation.

Atom Chemical Potential

As specifically corrected to fit experimental formation enthalpies, the formation enthalpy data from the Materials Project is implemented here to build $\Delta\mu_i$ constraints

$$\mu_i = \mu_i^{\mathrm{DFT}} + \Delta\mu_i^{\mathrm{MP}}$$

$$\sum n_i\mu_i = 1\mu_W - 1\mu_{TM} = (1\mu_{Wi} - 1\mu_{TMi})^{\mathrm{DFT}} - (1\Delta\mu_W - 1\Delta\mu_{TM})^{\mathrm{MP}}$$



Charge Correction

Supercell approach: The isolated charged defect → periodically repeated array of defects → artificial electrostatic interactions

The FNV scheme: $\Delta E^{\text{corr}} = E_{\text{model}}^{\text{iso}} - E_{\text{model}}^{\text{periodic}} + QC$

Image charge

- Determine the z-dependent dielectric function
- use a model charge distribution to emulate the behavior of the defect charge state, and find the model potential by solving the Poisson equation.

$$\rho(\mathbf{r}) = \frac{1}{\left(\sqrt{2\pi}\sigma\right)^3} e^{-\frac{(\mathbf{r}-\mathbf{r_0})^2}{2\sigma^2}} .$$

- Do it twice with periodic boundary conditions and zero boundary conditions. The electrostatic energy is given by:

$$U = \frac{1}{2} \int_\Omega \rho V$$

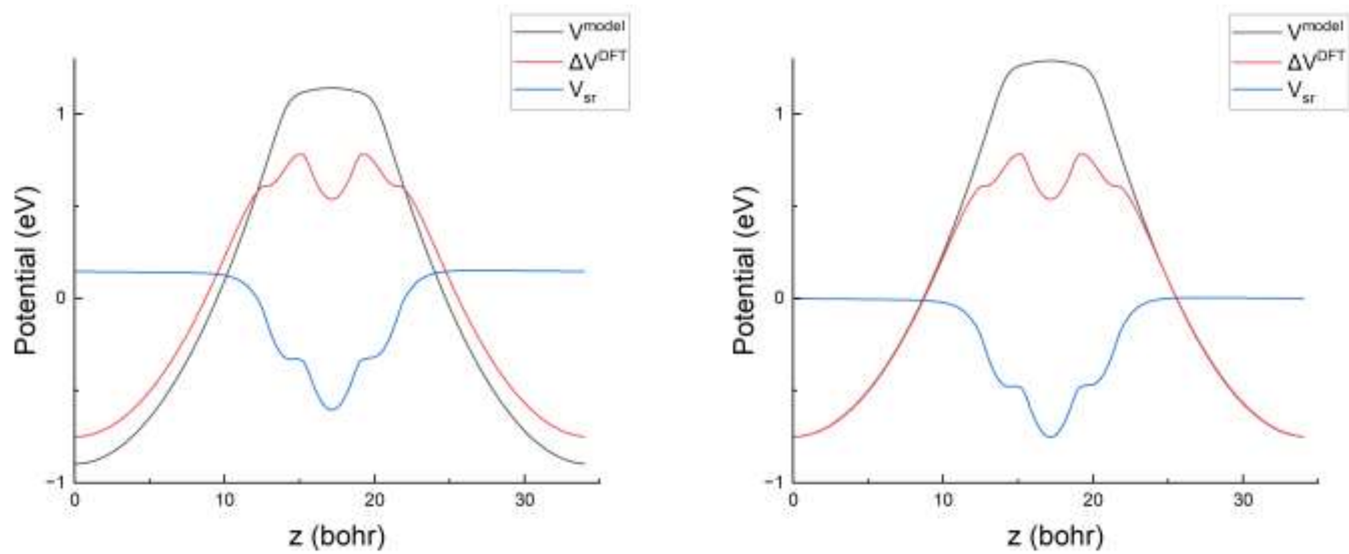
Charge Correction

Potential alignment:

Manually set C until potential alignment:

$$\Delta V^{DFT} - V^{model} - C \rightarrow 0 \qquad \text{(away form defect)}$$

Correction Energy: +1 : 0.094 eV -1 : 0.071 eV

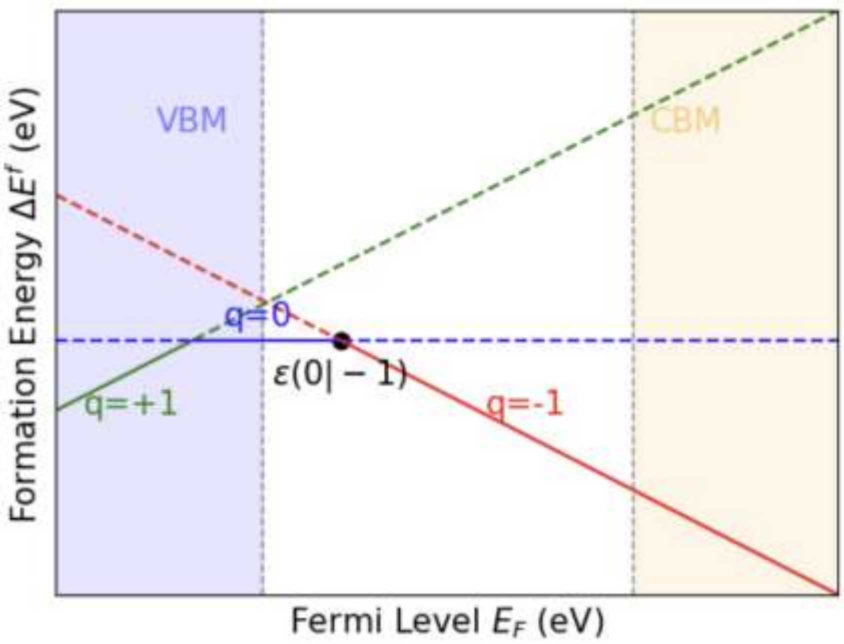


Formation Energy Diagram

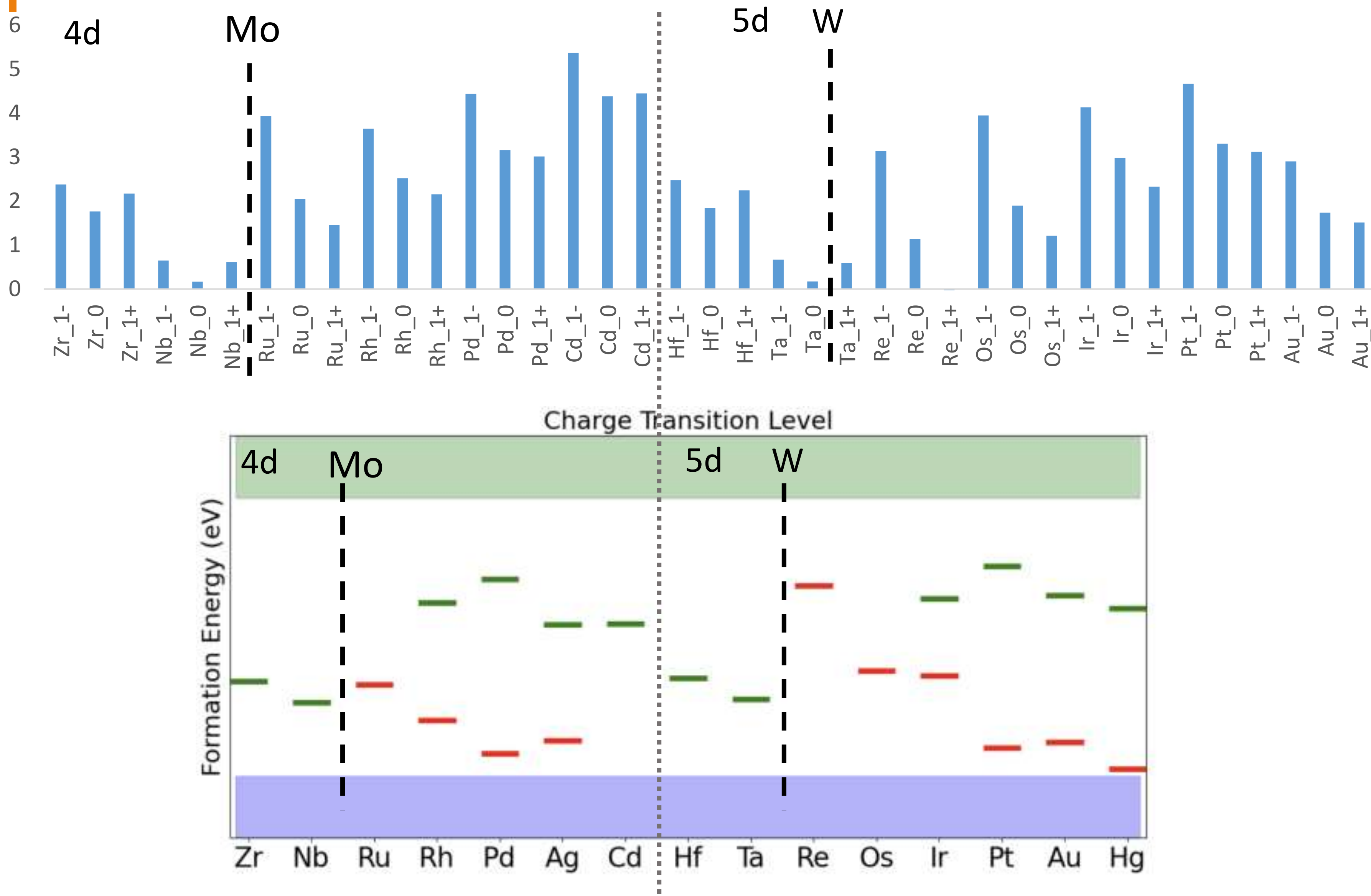
Table 3 Formation energy values of charge states +1, 0, -1.

Defect	q	$E_{D,q}$ - E_{Host}	qE_{VBM}	$\Sigma n_i \mu_i$ (S-rch/Wrich)	E_{corr}	Ef(E_F=VBM) (S-rch/W-rich)
Nb_W^{+1}	+1	4.076	-1.518	-2.001/-1.675	0.094	0.651/0,977
Nb_W^{0}	0	2.232	0	-2.001/-1.675	0	0.231/0.557
Nb_W^{-1}	-1	1.098	1.518	-2.001/-1.675	0.071	0.686/1.012

$$\sum n_i \mu_i = 1\mu_W - 1\mu_{Nb} =$$
$$= (\Delta\mu_W + \mu_W^0) - (\Delta\mu_{Nb} + \mu_{Nb}^0)$$
$$= \begin{cases} -2.001\text{eV} & \text{S rich} \\ -1.675\text{eV} & \text{W rich} \end{cases}$$

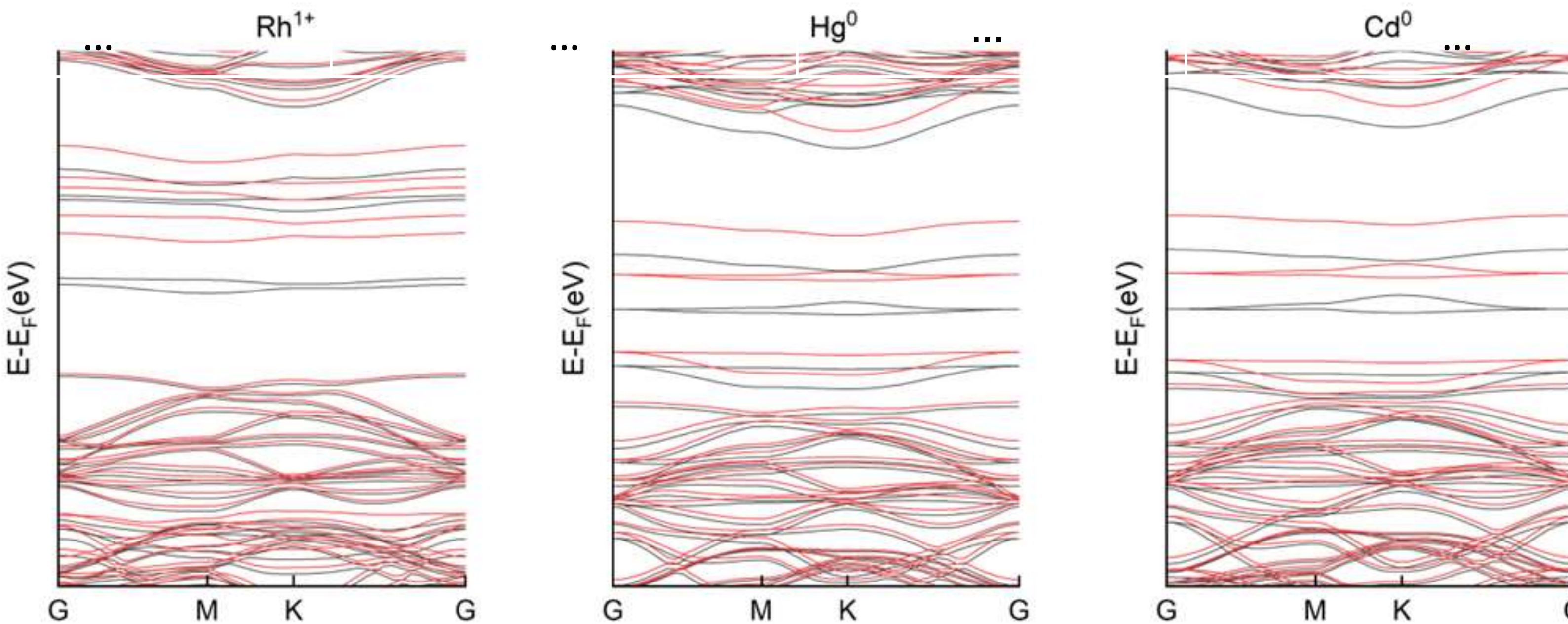


Formation Energies & Charge Transition Levels



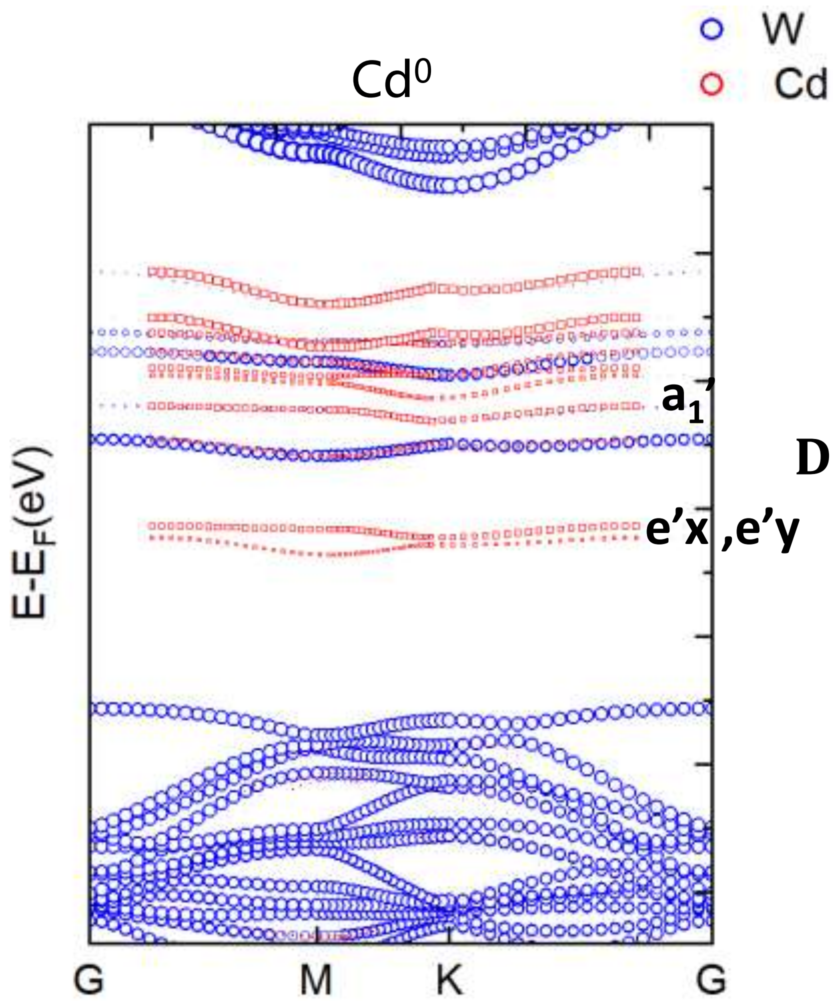
Spin states & Band structure

TM Defect	Charged state	Symmetry	ΔE (Singlet-Triplet)(eV)
Rh	+1	C_S	0.06
Hg	0	D_{3h}	0.04
Cd	0	D_{3h}	0.03
Ag	-1	C_{2v}	-0.05
…	…	…	…



Deep Center: $\Delta CB > k_BT$ and $\Delta VB > k_BT$

Projected Band structure



$\mathbf{D_{3h}}$: $\mathbf{A_1{}' \oplus E' \oplus E''}$

Outlook

Opportunities and Challenges in Complex Parameter Space

Defect Structure ✕ Charge State ✕ Spin State

Vacancies, Interstitials, Substitutional impurities;

Divacancies, Di-interstitial, Dopant-vacancy complexes, …;

……

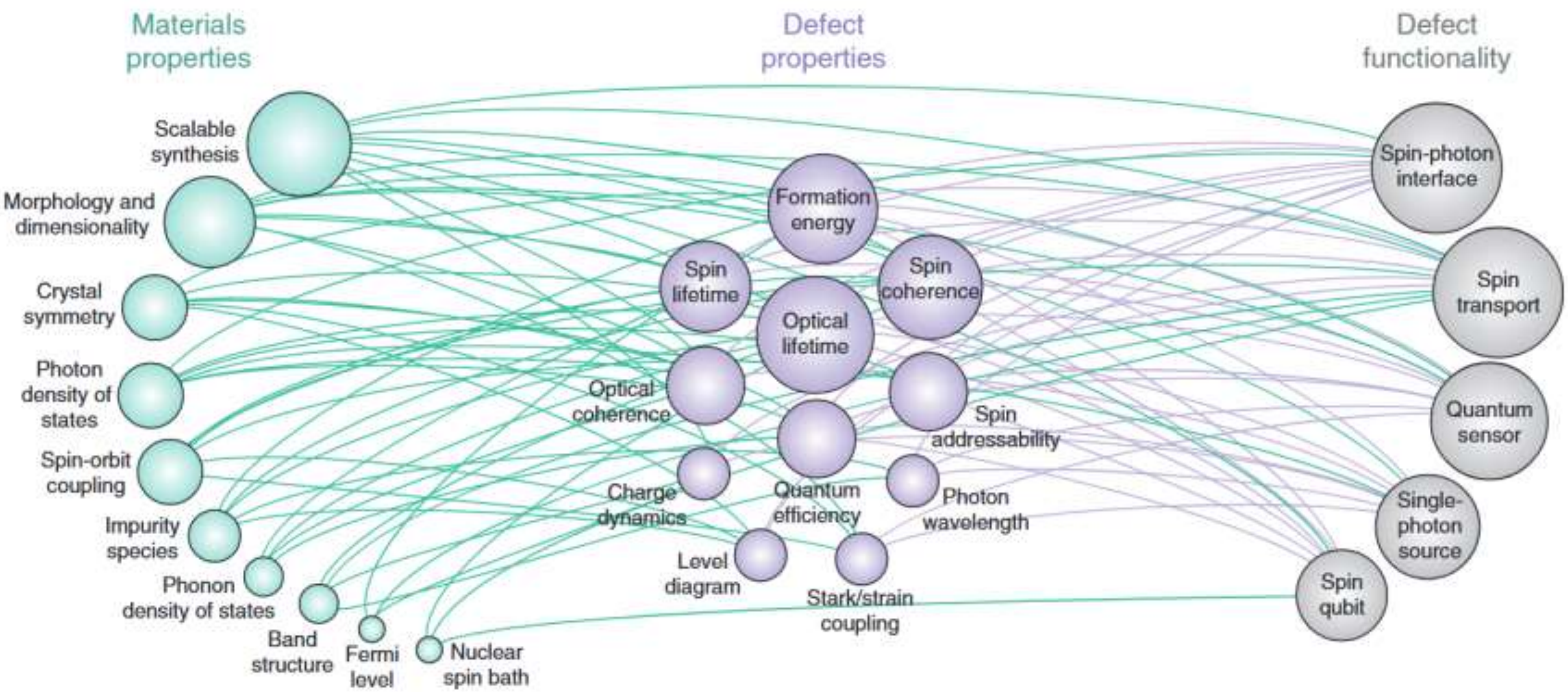
Known defects involve as many as **four** sites.

beyond traditional host materials

Millions of potential **host-defect combinations**.

Outlook

Network of property-function relationships



Outlook – Related Work

1. Machine Learning-Enabled Design of Point Defects in 2D Materials for Quantum and Neuromorphic Information Processing

Nathan C. Frey, Deji Akinwande, Deep Jariwala, and Vivek B. Shenoy
ACS Nano 2020 14 (10), 13406-13417

2. Sparse representation for machine learning the properties of defects in 2D materials.

Kazeev, N., Al-Maeeni, A.R., Romanov, I. et al.
npj Comput Mater 9, 113 (2023)

3. Identifying candidate hosts for quantum defects via data mining

Ferrenti, A.M., de Leon, N.P., Thompson, J.D. & Robert J. C.
npj Comput Mater 6, 126 (2020)